

CHARKOV, Russian Federation

WMO: 298640

Lat: 53.717N

Lon: 90.367E

Elev: 525

StdP: 95.17

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.7	-30.2	-35.5	0.1	-32.6	-33.0	0.2	-30.0	9.4	-5.7	8.4	-7.8	2.5	320	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	28.7	17.7	27.0	17.1	25.1	16.4	19.1	25.9	18.2	24.5	17.3	23.3	2.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.8	12.7	21.8	15.8	11.9	21.1	14.8	11.2	20.3	56.9	25.9	53.6	24.5	50.9	23.3	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.4	6.2	DB	-34.8	32.0	3.6	2.0	-37.4	33.4	-39.5	34.6	-41.5	35.7	-44.1	37.1	(4)
(5)			WB	-34.9	21.2	3.5	1.6	-37.4	22.4	-39.4	23.3	-41.3	24.2	-43.8	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.1	-18.1	-14.5	-5.5	3.1	9.6	16.7	18.0	15.5	8.7	1.0	-7.6	-14.7	(6)	
	DBStd	13.57	6.99	6.99	6.65	5.12	4.89	3.84	2.80	3.42	4.15	5.08	6.85	7.10	(7)	
	HDD10.0	3968	871	687	479	212	69	3	0	2	73	281	527	764	(8)	
	HDD18.3	6361	1129	920	737	457	274	74	41	100	291	538	777	1023	(9)	
	CDD10.0	716	0	0	0	4	56	202	247	173	33	2	0	0	(10)	
	CDD18.3	68	0	0	0	0	2	24	29	13	1	0	0	0	(11)	
	CDH23.3	804	0	0	0	1	45	319	266	159	14	0	0	0	(12)	
	CDH26.7	176	0	0	0	0	6	77	55	36	2	0	0	0	(13)	
	Wind	WSAvg	2.7	2.4	2.5	2.9	3.2	3.5	2.9	2.5	2.4	2.6	2.4	2.3	2.5	(14)
(15) Precipitation	PrecAvg	298	5	3	6	13	29	61	70	59	27	11	9	6	(15)	
	PrecMax	478	13	9	29	40	113	129	109	121	67	49	25	13	(16)	
	PrecMin	183	0	0	1	1	5	16	39	16	4	1	2	0	(17)	
	PrecStd	62	3	2	6	11	22	27	17	31	14	9	6	3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	2.9	13.1	21.1	27.7	31.4	30.8	29.7	25.4	17.8	9.1	1.6	(19)	
		MCWB	0.1	-0.6	6.0	11.0	14.7	17.8	18.7	19.3	15.6	9.8	4.9	-1.4	(20)	
	2%	DB	-1.9	0.1	9.0	17.8	24.4	28.8	28.3	27.3	21.1	13.6	6.1	-0.9	(21)	
		MCWB	-4.0	-2.5	3.3	9.2	13.3	16.9	18.3	18.1	13.8	8.1	2.8	-3.4	(22)	
	5%	DB	-5.0	-1.9	6.1	15.1	21.8	26.9	26.5	24.7	18.6	11.0	4.0	-2.7	(23)	
		MCWB	-6.8	-4.0	1.6	7.7	12.4	16.4	17.7	16.9	12.6	6.7	1.2	-4.9	(24)	
	10%	DB	-7.9	-4.4	4.0	12.3	19.3	25.0	24.5	22.6	16.4	8.8	1.8	-4.6	(25)	
		MCWB	-9.4	-6.2	0.3	6.1	11.3	15.7	17.0	16.2	11.2	5.1	-0.6	-6.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	-0.4	6.5	11.7	16.3	19.2	20.8	20.4	16.7	10.4	5.1	-1.1	(27)	
		MCDB	2.6	2.3	12.6	20.2	24.3	27.5	28.3	27.5	24.0	16.0	8.8	1.3	(28)	
	2%	WB	-3.7	-2.3	3.7	9.6	14.5	18.2	19.5	18.7	14.6	8.4	3.1	-3.3	(29)	
		MCDB	-1.9	0.1	8.4	16.8	22.5	25.9	26.3	25.0	19.8	13.4	6.0	-1.1	(30)	
	5%	WB	-6.7	-4.0	1.9	7.9	13.1	17.4	18.5	17.6	13.1	6.8	1.2	-4.9	(31)	
		MCDB	-5.0	-1.8	5.9	14.3	20.4	24.8	24.4	23.5	17.7	10.7	3.8	-2.9	(32)	
	10%	WB	-9.3	-6.2	0.3	6.4	11.6	16.3	17.6	16.6	11.7	5.2	-0.6	-6.3	(33)	
		MCDB	-8.0	-4.4	3.7	11.8	18.4	23.5	23.0	21.9	15.8	8.6	1.7	-4.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	9.5	11.1	11.6	12.0	12.0	10.3	10.9	10.9	9.8	8.2	7.9	(35)	
		MCDBR	9.6	10.6	13.6	16.7	16.6	15.6	13.8	15.2	15.2	13.1	9.5	8.8	(36)	
		MCWBR	8.2	8.5	9.0	9.6	8.5	6.5	6.1	7.2	8.7	8.2	6.7	7.3	(37)	
	5% WB	MCDBR	9.4	10.5	12.9	15.5	14.4	13.3	11.5	13.2	13.7	12.6	9.3	8.8	(38)	
		MCWBR	8.1	8.4	8.9	9.4	7.9	6.0	5.8	6.7	8.3	8.0	6.8	7.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.269	0.305	0.355	0.400	0.406	0.398	0.410	0.395	0.357	0.338	0.308	0.273	(40)	
	taud		2.087	2.047	2.013	2.131	2.188	2.310	2.298	2.311	2.356	2.279	2.134	2.043	(41)	
	Ebn at Noon		673	766	803	823	840	851	833	823	811	731	610	570	(42)	
	Edh at Noon		70	102	133	136	135	121	121	113	95	81	67	60	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.00	2.03	3.49	4.50	5.29	6.05	5.47	4.59	3.23	1.79	0.89	0.65	(44)	
	RadStd		0.08	0.15	0.29	0.25	0.52	0.50	0.34	0.36	0.30	0.17	0.10	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page